DO NOW — RECALL

Lesson 048 · C# List<T>: Add, Insert and Remove

Name: _____ Class: _____ Date: _____

1. What does Array.Length give?

2. What is zero-based indexing?

3. What is aliasing between references?

4. A rising-edge D-type flip-flop starts with Q=0. D rises to 1 and then returns to 0 entirely between two rising clock edges. Explain why Q remains 0.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Number of elements in the array.

2. Model answer

First item is at index zero.

3. Model answer

Two references identify the same object.

4. Model answer

No rising clock edge samples D while it is 1. The flip-flop holds Q=0 between active edges and samples 0 at the next rising edge.

By the end, you can…

• List<T> is a resizable sequence. Add appends; Insert inserts at a specified index and shifts following items.

• RemoveAt removes an item by index; Remove(value) removes the first matching value instead.

• Unlike an array, List<T> changes Count as items are inserted/removed. An index must still be within the valid bounds.

Vocabulary — know these terms

List<T> — Resizable generic sequence.

Insert — Adds at a specified index.

RemoveAt — Removes item by index.

Count — Number of items currently stored.

NEW LEARNING

Read, annotate and explain

Observe

List<T> is a resizable sequence. Add appends; Insert inserts at a specified index and shifts following items.

Explain

RemoveAt removes an item by index; Remove(value) removes the first matching value instead.

Apply

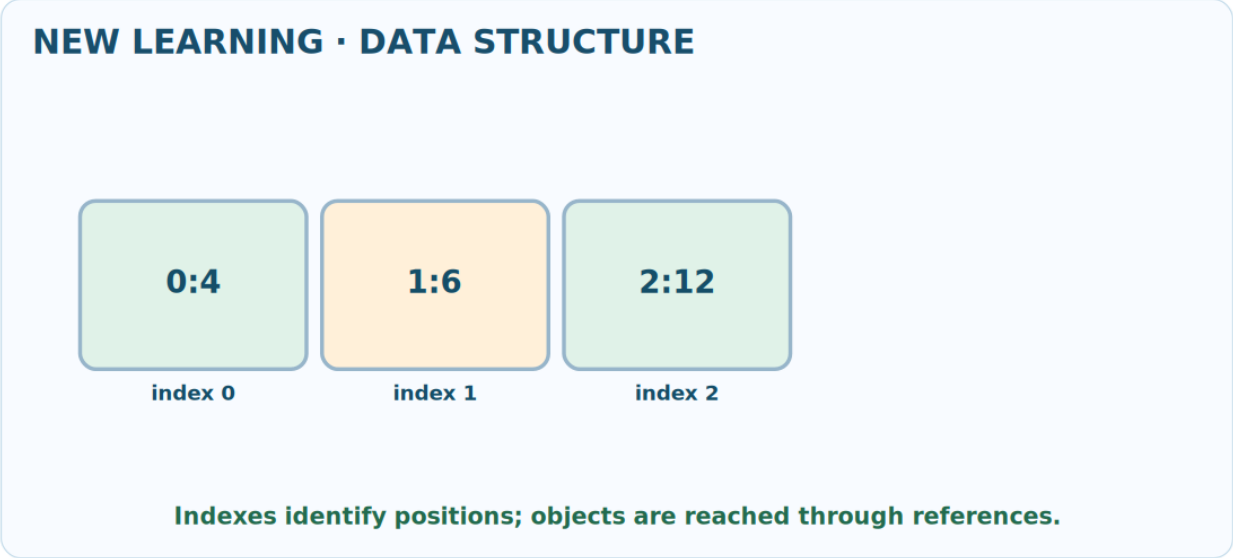
Unlike an array, List<T> changes Count as items are inserted/removed. An index must still be within the valid bounds.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



NEW LEARNING · DATA STRUCTURE: 0:4, 1:6, 2:12

I DO — WORKED EXAMPLE

Follow the worked method

Trace a list after inserting at the start, then removing the last element.

Trace and justify

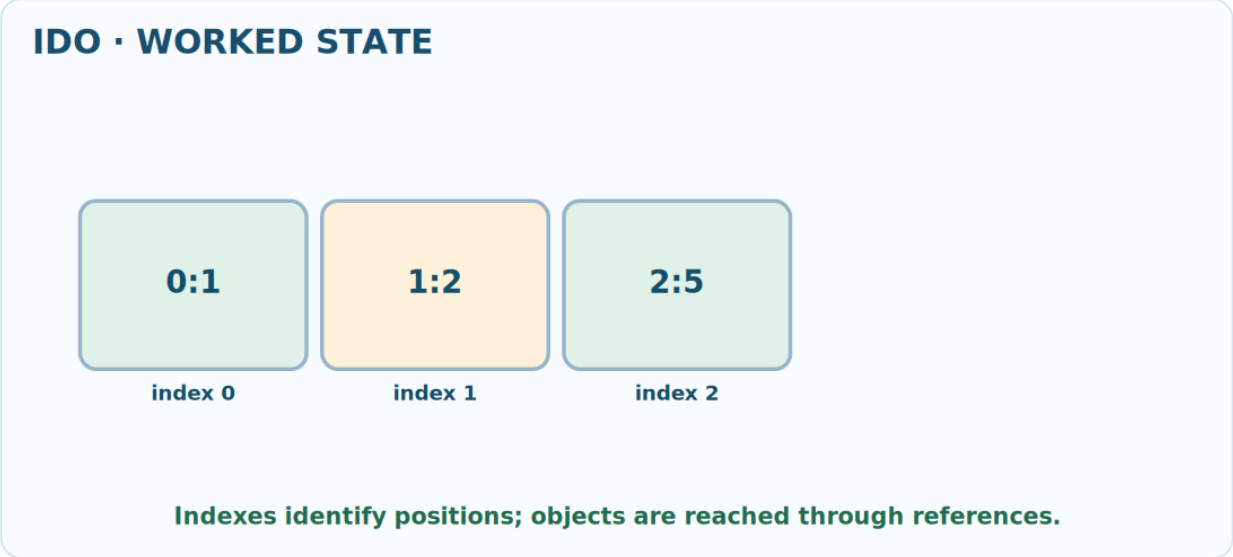
After Insert: [1,2,5,7]; RemoveAt(3) removes 7; final [1,2,5].

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace a list after inserting at the start, then removing the last element.

After Insert: [1,2,5,7]; RemoveAt(3) removes 7; final [1,2,5].



IDO · WORKED STATE: 0:1, 1:2, 2:5

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

A different list starts [3,6,9]. Add 12, then remove the VALUE 6. What remains?
Source code / input:

```
var n=new List<int>{3,6,9};
n.Add(12);
n.Remove(6);
```

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Trace Insert(1,6) then RemoveAt(2) for [4,8,12]. [6]

Question 2

Explain the difference between Remove(5) and RemoveAt(1) for [2,5,5,9]. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

For [1,2,3], write C# that inserts 0 at start, appends 4 and outputs Count; state why List<T> suits a growing collection. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

RemoveAt(5) means remove the value 5.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

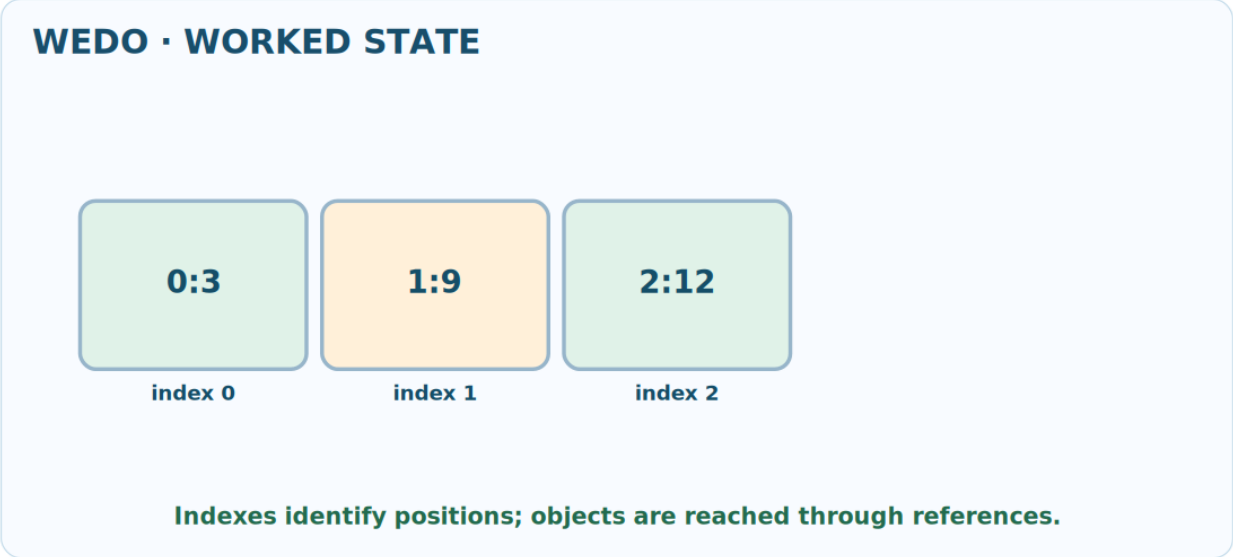
Trace and justify
[3,9,12]. Add appends 12; Remove(6) removes matching value 6, unlike RemoveAt(6).

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A different list starts [3,6,9]. Add 12, then remove the VALUE 6. What remains?

[3,9,12]. Add appends 12; Remove(6) removes matching value 6, unlike RemoveAt(6).



WEDO · WORKED STATE: 0:3, 1:9, 2:12